

Exercise 1: Holes in a conducting sphere

- a) Since the sphere is made of conducting material, the free charges rearrange to compensate the fields from q_a and q_b . These charges distribute themselves on the surfaces surrounding q_a and q_b with an opposite sign than the respective point charge. This means, q_a is surrounded by $-q_a$ and q_b is surrounded by $-q_b$. The surface of the sphere with radius r is $A = 4\pi r^2$, and for the surface charge density one therefore obtains $\sigma = \frac{q}{A}$. The surface charge densities are thus

$$\sigma_a = -\frac{q_a}{4\pi a^2}, \quad \sigma_b = -\frac{q_b}{4\pi b^2}. \quad (1)$$

The conducting sphere as a whole is neutral and the charge $q_a + q_b$ distribute itself uniformly on the outer spherical surface:

$$\sigma_R = \frac{q_a + q_b}{4\pi R^2}. \quad (2)$$

- b) The field of a sphere, outside of it, is indistinguishable from that of a point charge at its center:

$$\vec{E}_{\text{outside}}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_a + q_b}{r^2} \cdot \vec{e}_r, \quad (3)$$

with $\vec{r} = r\vec{e}_r$ and $\vec{e}_r$ being the radial unit vector from the center of the sphere.

- c) The fields within the holes correspond to the fields of the respective point charges:

$$\vec{E}_a = \frac{1}{4\pi\epsilon_0} \frac{q_a}{r_a^2} \cdot \vec{e}_a, \quad \vec{E}_b = \frac{1}{4\pi\epsilon_0} \frac{q_b}{r_b^2} \cdot \vec{e}_b, \quad (4)$$

with $\vec{r}_{a,b} = r_{a,b} \cdot \vec{e}_{a,b}$ and $\vec{e}_{a,b}$ being radial unit vectors from the centers of the holes. We must have $r_a < a$ and $r_b < b$.

- d) No, there is no such force between the two charges q_a and q_b . In the conducting sphere the charges move freely, until they completely compensate q_a and q_b .

In other words: The Coulomb force between q_a and q_b will be compensated by the opposite force from σ_b on q_a and from σ_a on q_b .

- e) If an additional charge is placed outside the conducting sphere, answers in task a) for σ_R and b) will change. The distribution of the outer surface charge changes, but not σ_a or σ_b . The force between q_a and q_b is still zero. The only electric field changing is $\vec{E}_{\text{outside}}$. Note that also the total charge on the surface $Q_R = \iint_{0,0}^{\pi,2\pi} \sigma_R \sin\theta R^2 d\theta d\phi$ does not change.

Exercise 2: Surface of a conductor

In electrostatics, the electric field $\vec{E}$ is zero in a conductor. If it wasn't zero, free charges inside the conductor would start moving along the field lines and we would be well outside the bounds of electrostatics.

- a) Consider an arbitrary point P on the surface of the conductor. We then place a very small circle C inside the conductor such that it is tangential to the surface of the conductor at P (see Fig. 2.1 bottom). With Stokes' Theorem and with $\vec{\nabla} \times \vec{E} = 0$ for every electric field in electrostatics we find

$$0 = \int_C (\vec{\nabla} \times \vec{E}) \cdot \vec{n} da = \int_{\partial C} \vec{E} \cdot d\vec{l}. \quad (5)$$

Since the electric field inside the conductor is zero, only the electric field on the surface of the conductor can contribute to the line integral along ∂C . If the direction of the border of the circle C at P is $\vec{l}$ we get in the limit of an infinitesimal circle

$$\vec{E}(P) \cdot \vec{l} = 0. \quad (6)$$

it follows that $\vec{E}(P)$ does not have a component that is tangential to its surface and $\vec{E}(P)$ must therefore stand perpendicular to the surface of the conductor. Since this is valid for any point P , it must be true on the whole surface.

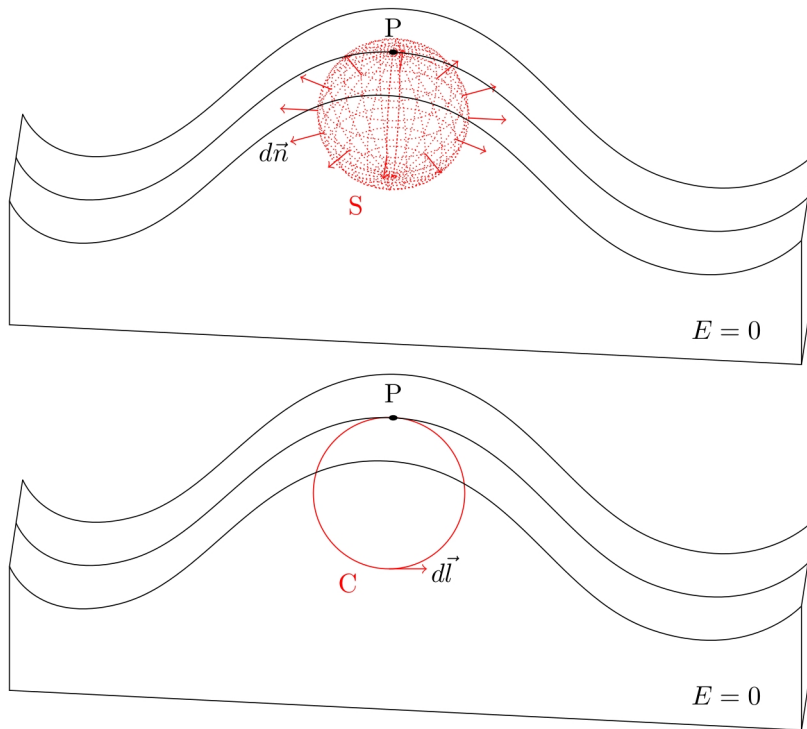


Abbildung 2.1: (Bottom) Circle inside the conductor tangential to P . (Top) Sphere inside the conductor tangential to P .

- b) Now place a very small sphere S inside the conductor tangential to the surface of the conductor at P . With Gauss' Theorem (and with its differential form $\vec{\nabla} \cdot \vec{E} = \rho/\epsilon_0$) we find

$$\int_{\partial S} \vec{E} \cdot \vec{n} da = \int_S \vec{\nabla} \cdot \vec{E}(\vec{r}) dr^3 = \frac{1}{\epsilon_0} \int_S \rho(\vec{r}) dr^3. \quad (7)$$

Since the electric field inside the conductor is zero and since the electric field at P has no tangential component to the surface, in the limit of an infinitesimal sphere S only the electric field at P contributes to the integral over the surface of S . It follows

$$\frac{dq(P)}{\epsilon_0} = \vec{E}(P) \cdot \vec{n} da \quad (8)$$

and equivalently

$$\frac{\sigma}{\epsilon_0} = \vec{E}(P) \cdot \vec{\hat{n}} = E(P). \quad (9)$$

Exercise 3: Electrostatics

a) We want to find the potential $\phi(x, y, z)$ that produces the given electric field:

$$-\vec{\nabla}\phi = \vec{E}(x, y, z) = \begin{pmatrix} ax^2 + bz \\ cy \\ bx \end{pmatrix}. \quad (10)$$

This means that

$$\frac{\partial\phi}{\partial x} = -ax^2 - bz, \quad \frac{\partial\phi}{\partial y} = -cy, \quad \frac{\partial\phi}{\partial z} = -bx. \quad (11)$$

From the third equation we get

$$\phi(x, y, z) = -bzx + f(x, y), \quad (12)$$

where $f(x, y)$ is a function that depends only on x and y . If we insert this into the second equation we obtain

$$\phi(x, y, z) = -bzx - \frac{c}{2}y^2 + g(x), \quad (13)$$

where $g(x)$ is only a function of x . From the first equation we finally get the potential

$$\phi(x, y, z) = -bzx - \frac{c}{2}y^2 - \frac{a}{3}x^3 + C, \quad (14)$$

where C is a constant.

b) The energy is calculated via the line integral

$$W = - \int_{P_1}^{P_2} \vec{F} \cdot d\vec{l} = -q \int_{P_1}^{P_2} \vec{E} \cdot d\vec{l} = +q[\phi(P_2) - \phi(P_1)] = -\frac{2a}{3}q. \quad (15)$$

c) The force on the electric dipole is

$$\vec{F} = (\vec{p} \cdot \vec{\nabla})\vec{E} = p_0(-\frac{\partial}{\partial x} + \frac{\partial}{\partial y})\vec{E} = p_0 \begin{pmatrix} -2ax \\ c \\ -b \end{pmatrix} \quad (16)$$

and the torque is

$$\vec{M} = \vec{p} \times \vec{E} = p_0 \begin{pmatrix} -1 \\ +1 \\ 0 \end{pmatrix} \times \begin{pmatrix} ax^2 + bz \\ cy \\ bx \end{pmatrix} = p_0 \begin{pmatrix} bx \\ bx \\ -ax^2 - bz - cy \end{pmatrix}. \quad (17)$$

The torque is zero when $x = 0$ and $z = -cy/b$. The force on $\vec{p}$ is never zero.

d) We can determine the charge density from the differential form of Gauss' law

$$\rho(x, y, z) = \epsilon_0 \vec{\nabla} \cdot \vec{E}(x, y, z) = \epsilon_0(2ax + c). \quad (18)$$

The total charge inside V is thus

$$Q = \int_V \rho dV = \epsilon_0 \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (2ax + c) dx dy dz = 4\epsilon_0 \int_{-1}^1 (2ax + c) dx = 8\epsilon_0 c. \quad (19)$$

e) The total energy of the electric field in V is

$$U = \int_V \frac{1}{2} \varepsilon_0 |\vec{E}|^2 dV \quad (20)$$

$$= \frac{1}{2} \varepsilon_0 \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 [(ax^2 + bz)^2 + (cy)^2 + (bx)^2] dx dy dz \quad (21)$$

$$= \frac{1}{2} \varepsilon_0 \left(\frac{8a^2}{5} + \frac{8b^2}{3} + \frac{8c^2}{3} + \frac{8b^2}{3} \right) \quad (22)$$

$$= 4\varepsilon_0 \left(\frac{a^2}{5} + \frac{2b^2 + c^2}{3} \right), \quad (23)$$

Note that only even terms of x, y, z (i.e. all terms of the form $\propto x^{2n}$ with $n = 0, 1, 2, \dots$) contribute to the integral.

Exercise 4: Laplace Operator

a) Note the polar coordinates

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} r \cos \phi \\ r \sin \phi \end{pmatrix} \quad (24)$$

or alternatively

$$\begin{pmatrix} r \\ \phi \end{pmatrix} = \begin{pmatrix} \sqrt{x^2 + y^2} \\ \tan^{-1}(\frac{y}{x}) \end{pmatrix} \quad (25)$$

With this, the following partial derivative is found:

$$\frac{\partial u}{\partial x} = \frac{\partial r}{\partial x} \frac{\partial u}{\partial r} + \frac{\partial \phi}{\partial x} \frac{\partial u}{\partial \phi} = \left(\cos \phi \frac{\partial}{\partial r} - \frac{1}{r} \sin \phi \frac{\partial}{\partial \phi} \right) u \quad (26)$$

The first equation comes from the chain rule while the second is a consequence of the following computations:

$$\frac{\partial r}{\partial x} = \frac{\partial \sqrt{x^2 + y^2}}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}} = \frac{r \cos(\phi)}{r} = \cos(\phi) \quad (27)$$

and

$$\frac{\partial \phi}{\partial x} = \frac{\partial \tan^{-1}(\frac{y}{x})}{\partial x} = \frac{-y}{x^2 + y^2} = -\frac{r \sin(\phi)}{r^2} = -\frac{\sin(\phi)}{r} \quad (28)$$

since the derivative of the inverse of the tangent is

$$(\tan^{-1}(x))' = \frac{1}{1 + x^2} \quad (29)$$

The derivative with respect to x and y can then be replaced by derivatives with respect to r and ϕ .

$$\frac{\partial}{\partial x} = \cos \phi \frac{\partial}{\partial r} - \frac{1}{r} \sin \phi \frac{\partial}{\partial \phi} \quad (30)$$

$$\frac{\partial}{\partial y} = \sin \phi \frac{\partial}{\partial r} + \frac{1}{r} \cos \phi \frac{\partial}{\partial \phi} \quad (31)$$

and directly as well the second derivatives (respect the chain rule):

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\cos \phi \frac{\partial u}{\partial r} - \frac{1}{r} \sin \phi \frac{\partial u}{\partial \phi} \right) \quad (32)$$

$$\frac{\partial^2}{\partial x^2} = \cos^2 \phi \frac{\partial^2}{\partial r^2} - \frac{\sin(2\phi)}{r} \frac{\partial^2}{\partial r \partial \phi} + \frac{\sin^2 \phi}{r^2} \frac{\partial^2}{\partial \phi^2} + \frac{\sin(2\phi)}{r^2} \frac{\partial}{\partial \phi} + \frac{\sin^2 \phi}{r} \frac{\partial}{\partial r} \quad (33)$$

$$\frac{\partial^2}{\partial y^2} = \sin^2 \phi \frac{\partial^2}{\partial r^2} + \frac{\sin(2\phi)}{r} \frac{\partial^2}{\partial r \partial \phi} + \frac{\cos^2 \phi}{r^2} \frac{\partial^2}{\partial \phi^2} - \frac{\sin(2\phi)}{r^2} \frac{\partial}{\partial \phi} + \frac{\cos^2 \phi}{r} \frac{\partial}{\partial r} \quad (34)$$

Then already

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \frac{\partial^2}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} + \frac{1}{r} \frac{\partial}{\partial r} \quad (35)$$

is found.

- b) As also the Laplace operator was calculated in two dimensions, we use the potential of a two dimensional dipole:

$$\Phi_{Dipol} = \frac{\vec{p} \cdot \vec{r}}{4\pi\epsilon_0 r^2} = \frac{p \cdot r \cos(\phi)}{4\pi\epsilon_0 r^2}, \quad (36)$$

with the angle between the dipole moment $\vec{p}$ and the position vector $\vec{r}$. Using these in the Laplace equation gives

$$\Delta \Phi_{Dipol} = \frac{\partial^2 \Phi_{Dipol}}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 \Phi_{Dipol}}{\partial \phi^2} + \frac{1}{r} \frac{\partial \Phi_{Dipol}}{\partial r} \quad (37)$$

$$= \frac{\partial^2}{\partial r^2} \left(\frac{p \cos(\phi)}{4\pi\epsilon_0 r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} \left(\frac{p \cos(\phi)}{4\pi\epsilon_0 r} \right) + \frac{1}{r} \frac{\partial}{\partial r} \left(\frac{p \cos(\phi)}{4\pi\epsilon_0 r} \right) \quad (38)$$

$$= 2 \cdot \left(\frac{p \cos(\phi)}{4\pi\epsilon_0 r^3} \right) - 2 \cdot \left(\frac{p \cos(\phi)}{4\pi\epsilon_0 r^3} \right) \quad (39)$$

$$= 0. \quad (40)$$

Exercise 5: Electron beam in an electric field

- a) Consider first the field that is produced by the bottom plate. The position of this plate along y is chosen such, that it lies in the origin of the coordinate system ($y = 0$). The electric field caused by this is zero along the x and z directions because of the problem's symmetry and it cannot depend on y , so that

$$\vec{E}_+ = E_+ \vec{e}_y \quad (41)$$

This is only true for the half space above the plate. The electric field in the space defined by $y < 0$ points down so that the field is given by

$$\vec{E}_+ = \begin{cases} E_+ \vec{e}_y & \text{if } y > 0 \\ -E_+ \vec{e}_y & \text{else.} \end{cases} \quad (42)$$

Consider now a cylindrical surface of length L and radius R , orthogonal to the plate (see Figure 2.9 in the lecture notes). The electrical flow is zero across the side of the cylinder so that

$$((E_+ \vec{e}_y) \cdot \vec{e}_y + (-E_+ \vec{e}_y) \cdot (-\vec{e}_y)) \pi R^2 = \frac{\sigma_+}{\epsilon_0} \pi R^2 \quad (43)$$

where we applied Gauss's theorem. The first term corresponds to the electric flow through the surface on top of the cylinder (normal vector equals $\vec{e}_y$) while the second one corresponds to the electric flow through the surface at the bottom of the cylinder (normal vector equals $-\vec{e}_y$). It follows,

$$E_+ = \frac{\sigma_+}{2\epsilon_0}. \quad (44)$$

Similarly, the electric field of the top plate (also independent of y in the spaces ($y > d$ and $y < d$)) is given by

$$\vec{E}_- = \begin{cases} E_- \vec{e}_y & \text{if } y > d \\ -E_- \vec{e}_y & \text{else.} \end{cases} \quad (45)$$

where

$$E_- = \frac{\sigma_-}{2\epsilon_0} = -\frac{\sigma_+}{2\epsilon_0} \quad (46)$$

The superposition of the fields gives $E = 0$ above the top plate and below the bottom one (E is the total electric field). The electric fields are added in the space between the plates.

$$\vec{E} = \begin{cases} \frac{\sigma_+}{\epsilon_0} \vec{e}_y & \text{if } y > 0 \text{ and } y < d \\ 0 & \text{else.} \end{cases} \quad (47)$$

b) The potential difference of the plates can be calculated from the electric field between them:

$$\begin{aligned} V &= - \int_{\text{Plate at potential 0}}^{\text{Plate at potential V}} \vec{E} \cdot d\vec{s} = - \int_{\text{Plate at potential 0}}^{\text{Plate at potential V}} E_y \vec{e}_y \cdot d\vec{s} \\ &= - \int_0^d -E_y ds = E_y \int_0^d ds = E_y d \end{aligned} \quad (48)$$

Because the potential difference V can be assumed to be positive, $d\vec{s}$ and $\vec{e}_y$ point in opposite directions, which explains the additional minus sign. Consider, that for negative V , $\vec{E}$ and $d\vec{s}$ point in the same direction and the additional minus sign drops out.

We can transform this to:

$$E = \frac{V}{d}, \quad (49)$$

where E in this case is the electric field in y -direction of the plate capacitor.

- c) First, we calculate the time of flight t_0 of the electron to cross the capacitor in x -direction. For that we calculate its velocity, by equating the electric energy gained by crossing the acceleration voltage and the kinetic energy:

$$E_{electr} = E_{kin} \quad (50)$$

$$\Leftrightarrow \frac{m}{2} v_0^2 = V_0 q \quad (51)$$

$$\Leftrightarrow v_0 = \sqrt{\frac{2V_0 q}{m}} . \quad (52)$$

From the velocity v_0 we get the time of flight t_0 :

$$v_0 = \frac{s}{t_0} \quad (53)$$

$$\Leftrightarrow t_0 = \frac{s}{v_0} = \frac{s\sqrt{m}}{\sqrt{2V_0 q}} . \quad (54)$$

From exercise b) we know, that for the plate capacitor, $E = V/d$ applies. The force acting on the electrons is:

$$F = E \cdot q = \frac{V \cdot q}{d} \quad (55)$$

So we get a constant acceleration in y -direction acting on the electrons in the capacitor

$$a = \frac{F}{m} = \frac{V \cdot q}{d \cdot m} . \quad (56)$$

This results in the deflection in y -direction

$$\Delta y = \frac{1}{2} a t_0^2 = \frac{1}{2} \frac{V q}{d m} \left(\frac{s\sqrt{m}}{\sqrt{2V_0 q}} \right)^2 \quad (57)$$

$$= \frac{V q s^2 m}{2 d m 2 V_0 q} = \frac{V s^2}{4 d V_0} \quad (58)$$

$$= \frac{50 \text{ V} \cdot (5 \text{ cm})^2}{4 \cdot 2 \text{ cm} \cdot 1000 \text{ V}} \quad (59)$$

$$\approx \underline{\underline{0.156 \text{ cm}}} . \quad (60)$$